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DFT Design Rule Checks

Learning Objectives

After completing this lab, you should be able to:

- Perform DRC checks on RTL, a pre-scan netlist, and a post-scan netlist
- Interpret the results of the DRC run from the `dft_drc` report
- Obtain an ATPG test coverage estimate



Lab Duration:
45 minutes

Lab 4

Overview

All files for this lab except for designs and libraries are located in the directory **lab4_drc**.

Directory Structure

lab4_drc	Current working directory
analyzed	Intermediate acs_read_hdl files
logs	Session log files
unmapped	Unmapped protocol
mapped	Gate-level netlist
mapped_scan	Will contain final netlist
reports	DFTC reports
tmax	Files for downstream tools
scripts	Constraint and run scripts
../ref/rtl/RISC_CORE/vhdl	Design files
../ref/db	Library db files

Answers & Solutions

This lab guide contains answers and solutions to all questions. If you need some help with answering a question, consult the back portion of this lab for help.

Instructions

During this lab, you will perform DFT design rule checks at various points in the scan insertion flow.

Task 1. RTL DFT DRC

1. Invoke DC/DFTC.

```
UNIX% cd lab4_drc
UNIX% dc_shell | tee logs/rtl_drc.log
```

2. Read the RTL design into DFTC and create the test protocol.

```
s scripts/task1_setup.tcl
```

Question 1. What command verifies that the RTL design and the protocol are compatible?

`dft_drc`

Question 2. What variable needs to be set to enable file and line number tracking for RTL DFT DRC?

```
1 # Enable file and line number tracking for RTL DFT DRC
2 # set hdlin_enable_rtl_drc_info true
3 set_app_var hdlin_enable_rtl_drc_info true
```

设置系统变量名不
建议使用set

3. Run dft_drc

```
dft_drc
```

4. Analyze the dft_drc report

Question 3. Were there any violations? What type?

```
DRC Report
Total violations: 401

001 PRE-DFT VIOLATIONS
018 Uncontrollable clock input of flip-flop violations (03)
90 DFF set/reset line not controlled violations (03)
1 Clock path affected by clock captured by clock 10 level sensitive clock_port violation (033)

Warning: Violations occurred during test design rule checking. (TEST-124)
Information: Test design rule checking completed. (TEST-123)
Warning: The current design contains unmapped components. The output netlist might not be read back into the system. (TEST-268)
```

建议start_gui打开
design vision看
schematic来debug

5. The TEST_MODE signal (active high) was not declared prior to creating the test protocol. Declare the signal with set_dft_signal and rerun dft_drc

```
set_dft_signal -view spec -port TEST_MODE -type Constant -active_state 1 -port TEST_MODE
remove_test_protocol
create_test_protocol
dft_drc
```

此处constant与
testmode效果完全
相同

Lab 4

Question 4. Are there any violations now?

```
-----  
DRC Report  
Total violations: 0  
-----
```

6. Exit dc_shell

```
exit
```

```
TCL:
dc_shell> set CLK_PORT [get_ports Clk]
{Clk}
dc_shell> echo $CLK_PORT
sell
dc_shell> get_object_name $CLK_PORT
Clk
```

Lab 4

Task 2. Pre-DFT DRC

1. Invoke DC/DFTC.

```
UNIX% dc_shell | tee logs/gate_drc.log
```

2. Read the design and perform a test-ready compile.

```
s scripts/task2_setup.tcl
```

Question 5. What command is used to perform a test-ready compile?

```
# Perform a test-ready compile
```

```
..... compile_ultra -scan
```

Question 6. From looking at the logfile, can you tell if automatic shift register identification was performed?

```
473 Information: Automatic shift-register identification is enabled for scan. Not all registers will be scan-replaced. (OPT-467)
```

若用 compile -scan
则不出现此信息

Question 7. What command verifies that the gate-level design and the protocol are compatible?

```
dft_drc
```

3. Run pre-DFT DRC

```
dft_drc
```

Question 8. Are the results the same as when dft_drc was run on the RTL? Is the report format the same?

No, there are more information about sequential cell report.

Enable enhanced dft_drc reporting and re-run dft_drc

```
set test_disable_enhanced_dft_drc_reporting FALSE
dft_drc
```

Question 9. What is the difference in the dft_drc report?

A list of report is given to show more information clearly.

DFT Design Rule Checks
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```
Summary of all DFT DRC violations
-----
0 TOTAL VIOLATIONS
Test Design rule checking did not find violations
-----
Sequential Cell Report:
Sequential Elements Detected: 261
Clock Gating Cells : 0
Synchronizing Cells : 0
Non-Scan Elements : 0
Excluded Scan Elements : 0
Violated Scan Elements : 0
(Traced) Scan Elements : 261 (100.0%) 0 ( 0.0%) 0 ( 0.0%)
-----
Information: Test design rule checking completed. (TEST-123)
```

Lab 4-5

Lab 4

Task 3. Post-DFT DRC

```
*****
Preview_dft report
For : 'Insert_dft' command
Design : RISC_CORE
Version: H-2013.03-SP5
Date : Mon Sep 10 15:55:41 2018
*****

Number of chains: 1
Scan methodology: full_scan
Scan style: multiplexed_flip_flop
Clock domain: mix_clocks
Scan enable: TEST_SE (no hookup pin)

Scan chain '1' (Instrn[0] --> Xecutng_Instrn[0]) contains 261 cells
```

1. Specify scan constraints and preview the scan architecture that will result.

```
s scripts/task3_setup.tcl
```

Question 10. How many scan chains does preview_dft report will be inserted?

1

2. Insert the scan chain(s).

```
insert_dft
```

Question 11. What command verifies that the inserted scan chain follows design rules?

```
dft_drc
```

3. Perform DFT DRC on the post-inserted netlist

```
dft_drc
```

```
-----
Traced Scan Chains
-----
Chain "1" traced through 261 flip-flops
```

Question 12. Were there any DRC violations?

0 total violations

```
-----
Traced Scan Chains
Chain "1" traced through 261 flip-flops
-----
Summary of all DFT DRC violations
-----
0 TOTAL VIOLATIONS
Test Design rule checking did not find violations

Sequential Cell Report: Sequential Cells Core Segments Core Segment Cells
Sequential Elements Detected: 261 0 0

Clock Gating Cells : 0
Synchronizing Cells : 0
Non-Scan Elements : 0
Excluded Scan Elements : 0
Violated Scan Elements : 0
(Traced) Scan Elements : 261 (100.0%) 0 ( 0.0%) 0 ( 0.0%)

Information: Test design rule checking completed. (TEST-123)
```

Question 13. What section of the dft_drc report is now included when performing DRC check on a post-scan inserted netlist?

```
dc_shell> dft_drc
In mode: Internal_scan...
Design has scan chains in this mode
Design is scan routed
Post-DFT DRC enabled
```

```
-----
Traced Scan Chains
-----
Chain "1" traced through 261 flip-flops
```

Question 14. What dft_drc command option is used to report an ATPG test coverage estimate after scan insertion?

```
dc_shell> dft_drc -help
Usage: dft_drc # Run Test Design Rule Checking
[-pre_dft] (Pre-DFT DRC only)
[-verbose] (Verbose messaging)
[-coverage_estimate] (estimate test coverage)
[-sample <sampld percentage of faults>]
(valid integer:
Range: 1 to 100)
dc_shell> dft_drc -coverage_estimate
```

4. Get a coverage estimate.

```
dft_drc ...
```

Question 15. What is the estimated coverage?

99.87%

Congratulations! You have completed the lab.

```

Information: Test design rule checking completed. (TEST-123)
Running test coverage estimation...
10276 faults were added to fault list.
ATPG performed for stuck fault model using internal pattern source.
-----
#patterns      #faults      #ATPG faults  test      process
stored        detect/active  red/au/abort  coverage  CPU time
-----
Begin deterministic ATPG: #uncollapsed_faults=8076, abort_limit=10...
0              5594    2482          0/0/0      75.73%     0.01
0              1086    1395          1/0/0      86.31%     0.02
0              550     845           1/0/0      91.67%     0.02
0              305     540           1/0/0      94.64%     0.02
0              193     345           3/0/0      96.54%     0.02
0              122     217           7/0/0      97.78%     0.02
0              82      132          10/0/0      98.61%     0.03
0              57      66           19/0/0      99.25%     0.03
0              50      7            24/0/2      99.82%     0.03
0              4       2            25/0/2      99.87%     0.03

Pattern Summary Report
-----
#internal patterns                                0
-----

Uncollapsed Stuck Fault Summary Report
-----
fault class      code      #faults
-----
Detected          DT          10224
Possibly detected PT           0
Undetectable      UD           39
ATPG untestable   AU           11
Not detected       ND           2
-----
total faults      10276
test coverage      99.87%
-----
Information: The test coverage above may be inferior
              than the real test coverage with customized
              protocol and test simulation library.

```